This example is functionally the same as the one prior to it, although it saves us some real estate on our screen and debatably makes the code easier to read. To say it another way, curly braces are useful for enclosing multiple statements in one continuous block of code, creating a compound statement while simple statements use the semicolon at the end of each single line. Remember that simple statements inside of compound statements still need to end with semicolons.

In these examples, you have also seen the application of the **parenthesis**, `( )`, to enclose arguments in a function or, as we will later see in more depth, for isolating arithmetic and comparison operations. Parentheses are used in pairs, so don't forget to close them off. The `loop()` function has nothing inside its parenthesis because no data is being passed to or from the function (more on this later). On the other hand, the example of `digitalWrite(9, HIGH)` passes two values to and from the `digitalWrite()` function, in this case the physical pin number on the interface board and its condition assignment. We have also used the parenthesis in separating a comparison operation in the case of `if (pin2 == HIGH)` where we are testing the condition of `pin2` against the constant `HIGH`. In arithmetic, parenthesis are also used to alter the order of operations as in `(2 +2) * 10`, although we will get to comparison and arithmetic operations in more detail momentarily.

Verifying and Uploading

Taking this concept of sketching to heart by tossing together some lines of code that we are still not fully up to speed on, we've got a sketch entered into the Arduino development environment and we think we have all the syntax and statements correct, but now what? In the last chapter, we blindy uploaded our first sketch full of confidence, but how do we know if the code we entered will work or not before uploading it to the Arduino board? To answer these questions, let's revisit uploading our sketch to the Arduino board, looking a little closer at the verifying and uploading process, including how to save your sketches and some of the common errors we might run into.

Verifying

As mentioned earlier, it is always a good idea to periodically verify a sketch that you're working on, just to make sure that you're on the right track, and even though this happens automatically before uploading a sketch to the Arduino board, it's a good idea to verify before uploading. To **verify** our code is to check that our code is correct and follows all the right syntactical rules that our sketches need to follow. We verify our sketch by hitting the Verify button in the top left of the toolbar in the Arduino development environment, as shown in Figure 2-4.